
Learnability, Reweighted: Which Tasks the Estimator Makes Active in Verifiable- Reward RL

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Abstract

Curriculum methods for RL with verifiable rewards summarize each task unit by a pass rate p and apply a learnability curve $f(p)$. That presumes the unit the curriculum scores is the random object the group estimator consumes. It need not be: a unit of uniformly uncertain tasks and a unit mixing mastered with impossible tasks share a mean pass rate, yet only the first yields mixed-outcome groups and non-zero updates. We formalize the mismatch. For a permutation-equivariant binary group estimator, activity is a functional of the success-count law, $\mathcal{A}(z) = \sum_k P(K=k | z) M_{\mathcal{E}}(k)$, with closed-form $M_{\mathcal{E}}$ for MaxRL, RLOO and GRPO; for practical MaxRL it is $2\{\Pr(K>0) - \mathbb{E}[K]/N\}$ under any joint law, and the familiar $2(\text{pass}@N - \text{pass}@1)$ is only its conditionally-i.i.d. atomic slice, which factors as $p(1-p) \cdot w_{N-1}(p)$ and recovers $p(1-p)$ at $N=2$. For a coarse unit, plugging in the mean pass rate over-predicts activity by exactly twice its excess all-fail probability. In a preregistered 48-block neural-maze intervention, replacing that plug-in with the shared-posterior count-law functional improves cov-AUC by $+0.00666$ (95% CI $[+0.00441, +0.00887]$), confirming causal relevance on that substrate. Where the scored unit is the estimator’s own, the deployed- N shape beats $p(1-p)$ by $+0.0480$ and reproduces on two further platforms using the same seeds. Three preregistered follow-ups bound it: performance rises past the deployed N , the score is starved as a standalone replay priority, and on a level aggregating heterogeneous tasks the original contrast reverses (-0.0032). The estimator defines the coefficient map; the curriculum defines the unit over which that map is averaged, and these operations do not commute. Coefficient activity is a principled, estimator-conditioned source of curriculum hypotheses, not a measure of learning utility.

1 Introduction

RL with verifiable rewards converts automatically checkable outcomes into policy updates, and every update first pays for a group of N model rollouts. Curricula make that budget productive by summarizing each task unit with a pass rate p and sampling by a learnability curve $f(p)$ (Tzannetos et al., 2023; Rutherford et al., 2024). That presumes the unit the curriculum scores is the random object the estimator consumes. Consider two units, each with mean pass rate $\frac{1}{2}$ at $N=16$ (Fig. 1). In the first, every task is solved about half the time: groups come back mixed and the estimator forms success-versus-failure contrast on nearly every one. In the second, half the tasks are mastered and half are impossible: every group is unanimous, all-pass groups carry no contrast, all-fail groups are dropped, and realized activity is exactly zero. Any $f(\bar{p})$ scores them identically; the estimator does not. The question is not which function of pass rate to use, but when pass rate is a sufficient statistic at all.

We answer for practical MaxRL. Its expected absolute coefficient mass obeys $A_N = 2(\Pr(K>0) - \mathbb{E}[K]/N)$ for any group law (Prop. 1); under conditionally i.i.d. rollouts this

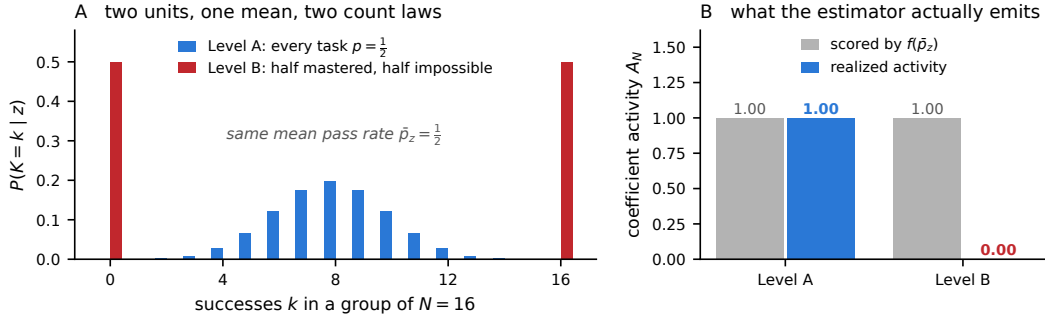


Figure 1: One mean pass rate, two opposite worlds. (A) Two curriculum units at $N=16$ with identical mean pass rate $\bar{p}_z = \frac{1}{2}$ and different success-count laws: Level A has every task at $p = \frac{1}{2}$; Level B mixes mastered with impossible tasks. (B) Scored by any $f(\bar{p}_z)$ both receive 1.00. Their realized coefficient activity $A_N = 2\{\Pr(K>0) - \mathbb{E}[K]/N\}$ is 1.00 and 0.00. Exact, not simulated.

is $2\{1 - (1-p)^N - p\} = 2(\text{pass}@N - \text{pass}@1)$ and factors as

$$u_N(p) = A_N(p)/2 = p(1-p) \cdot w_{N-1}(p),$$

where $w_T(p) = \sum_{k=1}^T (1-p)^{k-1}$ is MaxRL’s own weight function (Cor. 1). Coefficient activity is learnability reweighted by the deployed estimator. $p(1-p)$ is the $N=2$ slice, and the peak $p_N^* = 1 - N^{-1/(N-1)} \simeq \log N/N$ is derived, not tuned. MaxRL asked which objective more compute should buy; we ask which tasks that objective should be fed.

Our contributions are organized by what each is entitled to claim.

1. Theory, proved and machine-verified. Activity is a functional of the success-count law, not of mean pass rate: $\mathcal{A}_{\mathcal{E},N}(z) = \sum_k P(K=k | z) M_{\mathcal{E}}(k)$, with closed-form $M_{\mathcal{E}}$ for MaxRL, RLOO and GRPO. The familiar p -curves are its conditional-i.i.d. slice; Cor. 2 prices a coarse scored unit exactly; the bridge $\mathbb{E}[\hat{g} | x] = u_N(p)(\mu_+ - \mu_-)$ is why a coefficient surrogate is defensible; Lemma 1 fixes the deployed truncation at $T = N - 1$. All checked as code contracts against the deployed estimators.
2. Registered and confirmed. In a fixed Acrobot pool at deployed $N=16$, u_{16} beats $p(1-p)$ by $+0.048$ under a frozen rule and reproduces on two further platforms using the same seeds (§3.1).
3. Registered boundaries, reported as first-class results. Sweeping only the score exponent, performance rises past the deployed N — the harder-peaked shape helps, its peak location does not (§3.2). As a standalone replay priority on AMaze the score is starved: one Bernoulli per level visit against a critic read every timestep (§3.3). And where the curriculum scores a level aggregating heterogeneous tasks, the contrast reverses, -0.0032 with CI $[-0.0054, -0.0011]$ and effects at or above the registered SESOI ruled out — a sign consistent with Cor. 2, which predicts the calibration bias and not the endpoint (§3.4). Granularity, not scale, separates these studies.
4. Preregistered correction, confirmed. On the same MAZE-SCORE substrate, replacing the mean-pass-rate plug-in by the shared-posterior count-law functional improves cov-AUC by $+0.00666$ (95% CI $[+0.00441, +0.00887]$; §3.4). This establishes causal relevance of the correction there, not that Cor. 2 predicted the learning sign or that either arm beats $p(1-p)$. The LLM-scale interaction and Countdown mechanism remain open and appendix-only.

Throughout, curricula are evaluated with the estimator they ship.

The identity is an estimator-side diagnostic, not a theorem of learning progress. The paper is deliberately narrower than the project record: exploratory mechanisms, failed hypotheses, robotics pilots, gate studies, and implementation detail remain in the appendix and artifact.

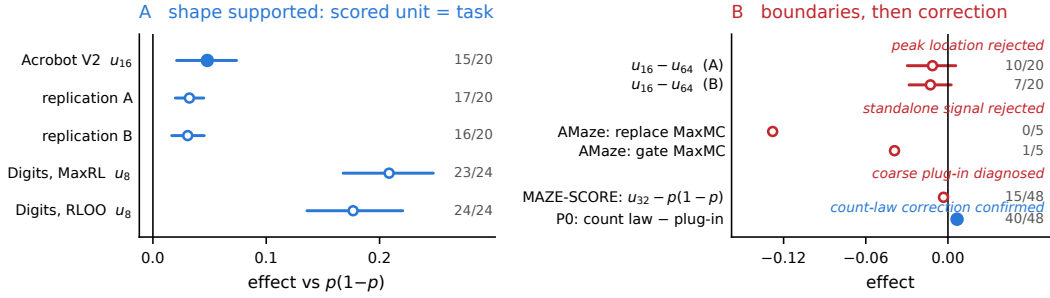


Figure 2: The claim scoreboard. (A) Every measured shape contrast against the $N=2$ slice, in studies whose scored unit is the i.i.d. unit; paired-bootstrap 95% intervals, positive-pair counts, filled marker for the one frozen primary. (B) Three preregistered boundaries plus the preregistered count-law correction on the same coarse-unit substrate. Rows there are different endpoints on their own studies' scales, shown together only to mark direction and interval; the filled blue marker is the correction's frozen primary. AMaze entries are five-seed development means at one-sixth budget and carry no interval.

2 Coefficient activity under practical MaxRL

Setup. For a task x let $\mathbf{r} \in \{0, 1\}^N$ be the group's rollout rewards, drawn from a joint law Q_x , and $K = \sum_i r_i$. Write $\bar{p} = \mathbb{E}_{Q_x}[K]/N$ and $q_N = \Pr_{Q_x}(K > 0)$; when the N rollouts are conditionally i.i.d. Bernoulli(p) these are p and $\text{pass}@N = 1 - (1-p)^N$. Let $s_i = \nabla_{\theta} \log \pi_{\theta}(y_i | x)$ be the response score and $\hat{g} = \sum_i w_i s_i$ the group update. Then $\text{pass}@N = 1 - (1-p)^N$. We analyze the zero-stabilizer idealization of the released practical centered MaxRL convention, which assigns

$$w_i = \frac{r_i}{K} - \frac{1}{N} \quad \text{when } K > 0, \quad w_i = 0 \quad \text{when } K = 0.$$

All-pass groups also have zero contrast. The released implementation uses a finite denominator stabilizer ε , multiplying the ideal coefficients by $K/(K + N\varepsilon)$ when $K > 0$; the identities below are exact for the zero- ε convention, not for its finite- ε magnitude.

Definition 1 (Finite-group activity geometry). A permutation-equivariant binary group estimator $\mathcal{E}$ maps a group outcome to coefficients whose absolute mass, under binary rewards, depends only on the success count: write $M_{\mathcal{E}}(k) = \|\mathbf{c}_{\mathcal{E}}(k)\|_1$. Its activity on a unit z whose groups follow the count law $P(K | z)$ is

$$\begin{aligned} \mathcal{A}_{\mathcal{E}, N}(z) &= \sum_{k=0}^N P(K=k | z) M_{\mathcal{E}}(k), \\ M_{\text{MaxRL}}(k) &= 2(1 - k/N) \mathbf{1}\{k > 0\}, \\ M_{\text{RLOO}}(k) &= \frac{2k(N-k)}{N(N-1)}, \\ M_{\text{GRPO}}(k) &= \frac{2}{N} \sqrt{\frac{N-1}{N}} \sqrt{k(N-k)}, \end{aligned}$$

the last for the sample-SD convention at zero stabilizer.

The sufficient statistic for estimator activity is the success-count law, not the mean pass rate. Under conditionally i.i.d. rollouts $P(K | z)$ is Binomial(N, p) and the sum collapses to a Bernstein polynomial in p — the familiar one-dimensional curves — but that collapse is an assumption about the scored unit, not a property of the estimator. All three masses above are verified against the deployed implementations for $N \in \{4, 8, 16, 32\}$ and every k (App. A). Choosing an estimator fixes $\{a_{\mathcal{E}, k}\}$ and therefore which tasks can emit an update at all, orthogonally to the objective-side question of which finite-rollout objective to deploy (Tajwar et al., 2026; Zheng, 2026). What singles out practical MaxRL is that its mass is affine in the two statistics $(q_N, \bar{p})$, so the sum collapses without any independence assumption.

Why ℓ_1 coefficient mass. Three properties, and one limit. On a live MaxRL group the coefficients sum to zero, so $\frac{1}{2}\|\mathbf{w}\|_1$ is simultaneously the total positive and the total negative

mass: it measures how much success-versus-failure contrast the estimator can form. If rollout scores are bounded, $\|s_i\| \leq B$, then $\|\sum_i w_i s_i\| \leq B\|\mathbf{w}\|_1$, so activity is a worst-case update envelope that depends on the estimator alone and not on unknown score geometry. And it is knowable before any rollout is spent, from a posterior over outcomes. The limit is equally sharp: activity is not the gradient norm and not expected improvement, because the realized update also carries the direction $\mu_+ - \mu_-$.

Lemma 1 (Practical truncation order). The expectation of the drop- $K = 0$ practical estimator equals the gradient of the $(N - 1)$ -term truncated maximum-likelihood objective, not the N -term objective.

For $J_T(p) = -\sum_{t=1}^T (1-p)^t/t$, direct conditioning gives

$$\mathbb{E}\hat{g} = \left(\frac{1 - (1-p)^N}{p} - (1-p)^{N-1} \right) \nabla p = \left(\sum_{j=0}^{N-2} (1-p)^j \right) \nabla p = \nabla J_{N-1}.$$

The $-(1-p)^{N-1}$ term is the price of an outcome-dependent baseline: the control variate is subtracted only on groups that happened to contain a success, so it is not mean-zero over the outcome law, and the shortfall is exactly the probability that a designated rollout is the group's lone success. The correction matters whenever a theoretical truncation parameter is mapped to a deployed group size. This $N - 1$ statement does not revise MaxRL's theorem for its uncentered conditional estimator; it characterizes the released centered convention after its all-fail group is zeroed, so the shift is a property of the deployed rule, not of the maximum-likelihood expansion. Details and the non-dropped control-variate convention are in App. A.

Proposition 1 (Exact coefficient mass under any group law). For any joint binary group law Q — no independence and no identical distribution required — the practical estimator's expected absolute coefficient mass is

$$A_N(Q) = \mathbb{E}_Q \left[\sum_{i=1}^N |w_i| \right] = 2(q_N - \bar{p}).$$

If the rollouts are conditionally i.i.d. Bernoulli(p) this is

$$A_N(p) = 2\{1 - p - (1-p)^N\} = 2(\text{pass@}N - \text{pass@}1).$$

We call A_N coefficient activity throughout; readers coming from the GRPO literature will know the same object as expected advantage mass. The realized mass of a group is the deterministic function $M(K) = 2(1 - K/N)\mathbf{1}\{K > 0\}$ — successes and failures each contribute $(N - K)/N$ when $0 < K < N$ — so $A_N(Q) = \mathbb{E}_Q[M(K)] = 2(q_N - \bar{p})$ needs no distributional assumption at all; only the scalarization to p does. This identity concerns coefficients, not gradient norm, signal-to-noise ratio, or expected improvement: score-gradient geometry can change its downstream effect.

The bridge to the expected update is exact: with $\mu_{\pm} = \mathbb{E}[s_i \mid r_i=1/0, x]$ we have $\mathbb{E}[\hat{g} \mid x] = u_N(p)(\mu_+ - \mu_-)$ for $u_N = A_N/2$, so u_N is the estimator-controlled scalar factor in the expected update and the conditional score geometry stays outside the mass calculation. Under i.i.d. rollouts $u_N(p) = \Pr(r_1=0, K > 0) = (1-p)\{1 - (1-p)^{N-1}\}$: activity is the probability that one rollout fails while a peer in the purchased group succeeds. Both dead zones follow — an all-fail task supplies no positive contrast, an all-success task no negative contrast — as does why N belongs in the curriculum's definition rather than its implementation.

Corollary 1 (Activity is reweighted learnability). Write MaxRL's population gradient as $\nabla J_T = \mathbb{E}_x[w_T(p)\nabla p]$ with weight function $w_T(p) = \sum_{k=1}^T (1-p)^{k-1}$ (Tajwar et al., 2026). Then, exactly,

$$u_N(p) = p(1-p) \cdot w_{N-1}(p).$$

By Lemma 1 the deployed estimator targets $T = N - 1$, and $p(1-p)\sum_{k=1}^{N-1}(1-p)^{k-1} = (1-p)\{1 - (1-p)^{N-1}\} = u_N(p)$. So coefficient activity is the learnability score reweighted

by MaxRL’s own weight function at the truncation the deployed estimator targets: $p(1-p)$ peaks at $\frac{1}{2}$ regardless of estimator, w_{N-1} decreases in p and grows with N as $p \rightarrow 0$, and their product peaks at $p_N^* = 1 - N^{-1/(N-1)}$, where an informative rollout meets an estimator that still cares. The familiar binary-success score $p(1-p)$ is both RLOO’s i.i.d. expected half-mass and practical MaxRL’s $N=2$ slice. SFL’s finite-group score is the corresponding realized RLOO mass up to a fixed factor; for practical MaxRL at arbitrary N , the estimator-matched curve is u_N .

Corollary 2 (Task granularity). Suppose a curriculum scores an aggregate z (a level, template, or domain) whose concrete tasks are $X \sim \nu(\cdot | z)$ with conditionally i.i.d. rollouts given X . Because $u_N''(p) = -N(N-1)(1-p)^{N-2} < 0$, plugging the aggregate’s mean pass rate $\bar{p}_z$ into u_N over-predicts its activity, and the gap is exactly the aggregate’s excess all-fail probability:

$$A_N(\bar{p}_z) - 2\mathbb{E}_X[u_N(p_X)] = 2[\Pr(K=0 | z) - (1 - \bar{p}_z)^N] \geq 0.$$

Both sides follow from Prop. 1 applied to the aggregate’s own group law, whose q_N is $1 - \Pr(K=0 | z)$. Curvature says who pays: near p_N^* the penalty is $\approx \frac{1}{2}|u_N''|\text{Var}(p_X)$ with $|u_N''(p_N^*)| = (N-1)/(1-p_N^*) \approx 34.7$ at $N=32$, against the exact and far milder $u_2(\bar{p}) - \mathbb{E}[u_2(p_X)] = \text{Var}(p_X)$. Raising N moves activity toward harder tasks and, near that active region, amplifies its sensitivity to how coarsely the curriculum names a task (§3.4). This is a local statement: for a fixed heterogeneous $\nu(\cdot | z)$ with all $p_X > 0$, both $\mathbb{E}[(1-p_X)^N]$ and $(1 - \bar{p}_z)^N$ vanish as $N \rightarrow \infty$, so the gap is not monotone in N globally.

Across estimators. The three masses differ in shape, zeros and tails rather than in a scale a learning rate could absorb: as $p \rightarrow 0$ MaxRL carries $(N-1)$ times RLOO’s, and sample-SD GRPO keeps relatively more on near-mastered prompts (Fig. 4B).

Exact-probability counter-test. The stronger test does not generalize the sampler mapping. A fresh 24-block Digits contextual bandit, computing each example’s exact correct-class probability before every update, leaves its frozen estimator-by-sampler interaction unsupported (+.016, CI $[-.017, +.047]$, $p = .350$; 15/24): MaxRL favors u_8 over $p(1-p)$ by +.208 (23/24), but so does RLOO (+.177, 24/24), reversing its prediction, and both matched samplers fall below uniform — the largest shape effect here and its sharpest negative in one study, under an internally hashed lock with no public pre-execution commit.

Instantiation. FrontierMax operationalizes the geometry without changing the within-group estimator: each task keeps discounted Beta evidence, draws a posterior pass rate $\tilde{p}_x$, and is sampled with probability $q(x) = \epsilon/M + (1-\epsilon)[u_N(\tilde{p}_x)]^\gamma / \sum_j [u_N(\tilde{p}_j)]^\gamma$. The floor ϵ , concentration γ , decay and posterior estimator remain empirical choices; the theory supplies the utility and its N -dependence, not a parameter-free scheduler. An exact deterministic posterior priority is also available (App. A), which the AMaze arm uses; scoring the posterior mean instead overstates activity by a closed-form uncertainty penalty — the same concavity that drives Cor. 2. Coefficient activity is the shared quantity, not a license to conflate a proved identity, a tested fixed-pool teacher and an untested replay overlay; Table 2 in App. B attaches a status to each object separately.

3 Evidence

Corollary 2 organizes what follows, and the organizing variable is not scale. The p -only reduction is exact when the curriculum’s scored unit is the unit over which rollouts are i.i.d., and lossy otherwise. Three studies satisfy that condition — the CPU chain scores a task with exact gradients, Acrobot scores a threshold predicate whose every rollout resets from a fresh seed, Digits scores one example whose exact class probability is recomputed each update — and in all three the deployed- N shape beats $p(1-p)$. MAZE-SCORE is the one study that violates it, drawing a single maze per group while the posterior pools over a level’s many mazes, and it is the one study where the shape loses. We read this post hoc, across studies not designed to test it. P0 then varied only the plug-in versus count-law functional and confirmed a learning gain on the same coarse-unit substrate (§3.4). Figure 2

states for each contrast what it is and is not: frozen primaries, supporting measurements, three preregistered boundaries, and the prospective correction.

3.1 Fresh fixed-pool Acrobot score test

We isolate the arbitrary- N score on official Gymnasium Acrobot-v1, where the scored unit is the task. Eight nested tip-height predicates share one task-blind H64 actor (640 parameters), the hardest being exactly native Acrobot success. All arms use practical MaxRL at $N = 16$, no hindsight, learning rate 3×10^{-4} , and a nominal two million paid transitions; twenty fresh paired seeds compare uniform sampling, $p(1 - p) = u_2$, and u_{16} under identical discounted Beta tracking. The internally frozen primary is target-uniform normalized transition-AUC for $u_{16} - p(1 - p)$, requiring a positive exact two-sided sign-flip test at .05 and a point estimate of at least +.01.

Table 1: Fresh Acrobot tournament. Paired differences use 20 fresh seeds under the source/runtime-locked protocol. The two uniform comparisons form one Holm family.

contrast	mean Δ	95% boot. CI	exact p	Holm p	decision
$u_{16} - p(1 - p)$	+.0480	[+.0209,+.0738]	.0034	—	supported by rule
$p(1 - p)$ —uniform	-.0062	[-.0226,+.0122]	.5078	.5078	not supported
u_{16} —uniform	+.0419	[+.0218,+.0606]	.0008	.0016	supported

Arm means are .64523 (uniform), .63907 ($p(1 - p)$), .68711 (u_{16}). The primary clears both frozen criteria with 15/20 positive pairs, supporting the deployed- N shape against its $N=2$ ablation in this fixed family — while $p(1 - p)$ itself is not supported over uniform by the frozen secondary.

3.2 Does the deployed- N peak location matter?

The peak p_N^* falls $.500 \rightarrow .169 \rightarrow .064$ for $N = 2, 16, 64$, so in every study above the winning arm is also the harder-peaked arm: “harder peaks beat $p(1 - p)$ ” and “the deployed- N peak is correct” predict identically there, and only the second would advance on the $p(1 - p)$ literature. Two preregistered campaigns separate them, holding the estimator at $N = 16$ in every arm and varying only the sampling exponent, analyzed once each.

An over-shooting arm scored by u_{64} does not lose to u_{16} : the paired difference $u_{16} - u_{64}$ is $-.0113$ (95% CI $[-.0297, +.0057]$, exact $p = .235$, 10/20) on one platform and $-.0128$ ($[-.0284, +.0024]$, $p = .131$, 7/20) on a second. A seven-point exponent sweep $N_{\text{score}} \in \{2, 4, 8, 16, 32, 64, 128\}$ gives arm means .644, .668, .665, .675, .679, .688, .681, with argmax at u_{64} , not at the deployed u_{16} , and Spearman rank correlation +.93 between exponent and mean. The curve rises past the deployed N : peak-location specificity is not supported at this operating point.

The same campaigns reproduce the shape effect on the shared seeds. Against $p(1 - p)$, u_{16} wins by +.0322 ($[+.0198, +.0449]$, exact $p = .000105$, 17/20) and +.0307 ($[+.0166, +.0453]$, $p = .000507$, 16/20), versus +.0480 in V2 — the first reproductions outside the machine that produced V2, both with a smaller p -value than V2’s own. Magnitude is platform-sensitive (about 35% attenuation on identical seeds between architectures), so we attribute each number to its platform. Because the two new campaigns share seeds and a deterministic engine, their agreement is a portability check, not additional independent seeds.

Why the curve rises past the deployed N . Branching an exact-gradient pool to measure true continuation utility $U_H(x) = J(\text{Train}_H(\theta; x)) - J(\theta)$ at $H = 8$, u_{16} ranks it far better than $p(1 - p)$ (+.106, 10 seeds, exact $p = .002$), while under RLOO — whose mass is $2p(1 - p)$ — the two are indistinguishable (+.0001). But U_H peaks near $p \approx .02$, below both p_{16}^* and p_{64}^* : activity ranks utility well and locates it poorly. Local activity and downstream benefit are also reported to diverge across domains under GRPO (Yang et al., 2026). Mechanism on 36 synthetic tasks, not evidence for a sampler.

The contrast stays descriptively positive on sampled-group (+.05294) and optimizer-update (+.05366) axes, and u_{16} receives 2.40 fewer groups and 3.42 fewer updates per million transitions, so it did not win by receiving more updates; these non-confirmatory axes identify no causal mechanism. Native-success transition-AUC is .31085, .30521, .37615 across arms. App. C reports the near-equal observed mass means and the audited but RNG-confounded precursor.

3.3 Where the score is starved: activity as a PLR priority on AMaze

The natural competitive test drops the activity priority into minimax’s robust PLR (Jiang et al., 2021; 2023) in place of MaxMC, running the shipped maze configurations verbatim. Our development sweep — seeds disjoint from any confirmatory block, 5,000 of the 30,000 shipped updates, five paired seeds per arm — is negative, and the reason is informative. Upstream robust PLR scores .629 and the no-curriculum control .539, with every activity variant below the control: replacing MaxMC with $\mathbb{E}[u_N(p) \mid \mathcal{D}]$ at the shipped 32×1 batch scores .500 ($N=8$; 0/5 seeds beat MaxMC, exact $p=.0625$) and .551 ($N=16$).

The cause is information, not shape. MaxMC reads critic disagreement $\max\text{-return} - V(s_t)$ at every timestep, with memory of the best return ever achieved on that level, while at the shipped `n_eval=1` a success-rate score gets one terminal Bernoulli per level visit — a two-valued posterior, nearly all ties under rank prioritisation — and `max_episode_steps=250` against a 256-step rollout forbids a second episode. Gating MaxMC by activity rather than replacing it recovers most of the gap (.590; $-.039$ to upstream, $1/5$) without clearing it. The AMaze student is PPO with GAE and never forms binary rollout groups, so the faithful mapping is outside this evidence.

3.4 A coarser task unit: mean pass rate is insufficient

Every positive above is at 640 or 4,810 parameters with exact or near-exact gradients. We preregistered the same contrast at neural scale: a 1.26M-parameter transformer on procedural 17×17 mazes across 13 goal-distance levels, estimator fixed to practical MaxRL at the deployed $N = 32$, three samplers (u_{32} , $p(1 - p)$, uniform) sharing one Beta posterior, decay, floor, level set, and per-block SFT checkpoint. Forty-eight paired blocks; endpoint mean pass@8 coverage over ten RL evaluations minus its post-SFT value; support required Holm $p < .05$, a positive bootstrap lower bound, and at least $+.005$.

The primary is practically ruled out by the frozen rule: u_{32} minus $p(1 - p)$ is $-.00324$ (95% CI $[-.00543, -.00111]$, exact $p = .0054$, 15/48), the interval’s upper bound falling below the SESOI. The registered secondary is supported (u_{32} beats uniform by $+.00888$, $[+.00657, +.01115]$, 41/48), so curriculum sampling helps here; what fails is that the rollout-aware shape is the better one. Per the branch fixed in advance, effects at or above the SESOI are ruled out under this protocol and no universal non-transfer is claimed.

Why: the curriculum names a coarser unit than the theory. The trainer draws one concrete maze per group and repeats that prompt N times, while the teacher’s posterior pools over a level’s many mazes: the i.i.d. unit is the maze, the scored unit is the level. Prop. 1 still holds exactly — what fails is its p -only reduction, which Cor. 2 prices. The telemetry confirms both identities to floating point across 41,101 (seed, arm, level, window) cells, each to $< 5 \times 10^{-16}$ and unchanged at window widths 10, 25, 50 (App. C). The gap is large where the score aims: at $\hat{p} \approx .11$ the binomial predicts 2.2% silent groups and 51.2% are silent (Fig. 5). Clustered on the 48 seed blocks, u_{32} realizes .580 [.570, .590] of its prediction against .703 [.691, .715] for $p(1 - p)$ (paired $-.123$, 48/48 blocks negative), losing 60.8% of its groups to silence against 32.1%. Corollary 2 predicts this activity-calibration gap, not either learning sign; the diagnosis is post-hoc.

Prospective correction. P0 then held the estimator, N , warmstart, task generator, budget, posterior, decay, floor and 48 paired seeds fixed, varying only the score applied to that posterior: the i.i.d.-at-the-mean plug-in versus realized count-law activity. The delivered intervention (mean visit TV .33597; frozen threshold .05) improves cov-AUC by $+.00666$

(95% CI $[+.00441, +.00887]$, exact paired $p = 9.56 \times 10^{-7}$, 40/48 positive blocks), exceeding the preregistered $+.005$ SESOI. The frozen verdict is supported: the correction is causally relevant on this substrate. This does not show that the algebra predicted the learning sign, that the correction alone explains the earlier contrast, or that either arm beats $p(1 - p)$.

Secondary records. The maze factorial, N -sweep, paid-probe, creation-channel and Count-down records remain in App. C; none supports named-method superiority. Their measurement lesson is to retain raw per-task outcomes so mean accuracy can be separated from standard $\text{pass}@k$.

Takeaway. If you are deploying this. (i) Score with u_M for $M \geq$ your deployed N : the deployed N is a floor on the score exponent, not an optimum (§3.2). (ii) Score the unit your estimator consumes; if the curriculum names a coarser bucket, score it by the realized count law $\hat{q}_z - \hat{p}_z$ rather than by $u_N(\hat{p}_z)$. Both come free from group outcomes you already log; the preregistered MAZE-SCORE correction confirms this on one substrate. (iii) Do not expect activity to replace a per-timestep regret signal; at one Bernoulli per visit it is a gate on a richer signal, not a signal (§3.3). (iv) Treat relabeling as a separate creation channel; its appendix-only gate record does not support a deployment prescription here (App. C). (v) Log the dead-group fraction and realized-versus-predicted mass beside $\text{pass}@k$: those two diagnosed §3.4.

4 Related work

Objective-side work such as MaxRL (Tajwar et al., 2026) and RL2ML (Zheng, 2026) asks which population or finite-rollout objective a group estimator implements, holding the task distribution fixed. Learnability curricula — ProCuRL (Tzannetos et al., 2023), SFL (Rutherford et al., 2024), LILO (Foster et al., 2025) — adapt that distribution using success variability; advantage-bandit curricula SEC (Chen et al., 2025) and DUMP (Wang et al., 2025) estimate activity empirically from observed advantages; TAC (Yang et al., 2026) adds first-order cross-domain transferability under GRPO. We study a complementary object: the finite-group activity geometry a deployed estimator induces (Def. 1), derived from its algebra before any curriculum hypothesis is tested. Superiority over the methods below is untested and not claimed.

Concurrent estimator and selection analyses. Closed-form magnitudes for the standard-deviation-normalized branch peak at $p = .5$ independent of N (Bay, 2026; Gu et al., 2026); practical-MaxRL mass instead peaks as $\ln N/N$. LZE’s LLM-scale study derives approximate $p(1 - p)G/n$ GRPO gradient variance under atomic i.i.d., fixed-baseline and homogeneous-score assumptions, then adds history and momentum (Cui, 2026); we derive exact coefficient activity under a count law, not gradient variance. F-GRPO treats rare correct modes omitted within a group by changing its weighting (Plyusov et al., 2026); we change task selection and expose coarse pooling. MoPPS establishes streaming Beta/Thompson selection (Qu et al., 2026); posterior machinery is precedent, while count-law sufficiency is our question.

Curricula and rollout allocation. Intermediate-difficulty sampling predates this work: ProCuRL’s proximal score reduces to $p(1 - p)$ under its optimal-success assumption (Tzannetos et al., 2023), and LILO (Foster et al., 2025) prioritizes binary reward variance. More tightly, SFL’s realized score $(k/N)(1 - k/N)$ (Rutherford et al., 2024) is exactly $M_{\text{RLOO}}(k)(N - 1)/(2N)$ at fixed N : our framework recovers SFL as the count-law curriculum for RLOO rather than replacing it as a heuristic. The surviving distinction is estimator-specific geometry (MaxRL peaks at $k = 1$, RLOO/SFL at the middle), bias from pooling a nonlinear score over a coarse unit, variable- N semantics (the fixed- N rescaling then varies), and scoring cost. For practical MaxRL, $u_N = A_N/2$ is its arbitrary- N expected half-mass. VCRL (Jiang et al., 2025) thresholds realized group variance; PLR (Jiang et al., 2021) and ALP-GMM (Portelas et al., 2020) use TD error, staleness or learning progress and so react to forgetting where instantaneous mass cannot — our AMaze arm is robust PLR (Jiang et al., 2023) with only the replay priority replaced (§3.3) — and generator-based UED (Dennis et al., 2020; Parker-Holder et al., 2022) creates tasks rather than pricing a fixed

pool. Prompt selectors (Zheng et al., 2025; Mao et al., 2026), per-prompt rollout counts (Nguyen et al., 2026) and adaptive allocators (Wang et al., 2026; Jiang et al., 2026; Yang et al., 2025; Xiong et al., 2025) act after the task distribution is given; we choose that distribution before generation at fixed N . HORA’s posterior-integrated hit utility makes the move Cor. 2 formalizes — integrate a nonlinear rollout utility over the relevant distribution rather than evaluate it at a mean — but for epistemic uncertainty about one prompt, where ours is heterogeneity inside a scored aggregate. BBG (Yuan et al., 2026) and boundary-aware curricula (Cai et al., 2026) target hit probability or boundary metrics, not coefficient mass. RL2ML (Zheng, 2026) separates finite-rollout objectives, update scale, variance and metric gain; its $T = N$ alignment retains $K = 0$ via the direct estimator or a score-baseline control variate, while our $T = N - 1$ statement is for the deployed hybrid that centers only when $K > 0$.

Coverage measurement. RLVR can improve pass@1 while reducing high- k coverage (Yue et al., 2025; Yuan et al., 2026; Zhou et al., 2026); entropy collapse is complementary evidence (Cui et al., 2025) and pass@ k objectives address the metric directly (Chen et al., 2025). We ask whether an external curriculum changes that coverage under two estimators. Relabeling and subproblem decomposition (Jiang et al., 2026) sit outside the allocation ceiling of Fig. 4A: at $A_N = 0$ no priority rule helps, and only creation adds signal.

5 Limitations

Coefficient mass ignores score-gradient direction, parameter sharing, optimizer state, task transfer, and non-i.i.d. rollouts: it is a sampler-facing surrogate, not a learning guarantee. The N -sweep is post-guidance on one co-designed synthetic chain; Digits is a contextual bandit. Acrobot uses one fixed eight-threshold family, a 640-parameter learner, no held-out task distribution or native named-method implementation, and 20 historically inherited rather than prospectively powered seeds; its exact test assumes sign exchangeability. The maze factorial uses one small transformer, one family and common learning rates; its easy-band mechanism is descriptive. The four families froze $\alpha = .05$ primaries separately, so no paper-level error rate is claimed; retained evaluations omit per-episode trajectories.

MAZE-SCORE bounds one task family, architecture, N and budget; because it also changes capacity, approximation, heterogeneity and granularity, it does not isolate scale. The post-hoc diagnosis exactly accounts for coefficient-mass calibration, not sole endpoint mediation, and its windows mix heterogeneity with learner nonstationarity. P0 isolates the score functional and confirms a $+.00666$ gain on the same task family and budget, but does not establish transfer, compare either arm with $p(1 - p)$, or show that the correction alone mediates the earlier contrast. The PLR study is a development negative, not evidence of superiority to PLR, robust PLR, ACCEL or another minimax baseline. No robotics performance is reported; paid-probe, recycling, Countdown, gate and transfer records remain secondary (App. C).

Registration timing remains a gap: the internally hashed Digits and Acrobot locks lack immutable public pre-execution commits, and the external maze wave-2 record has no vendored locking commit. Maze checkpoint trajectories remain external and the Digits replay payload has no download URI; App. D itemizes these gaps and the 54-row compact registry. They limit auditing and preclude a broad neural claim.

6 Conclusion

Estimator activity is a functional of the success-count law. Where the curriculum scores the estimator’s own unit, the deployed- N shape beat both u_2 and uniform sampling, reproduced twice with the same seeds; three preregistered follow-ups then bounded it. On the coarse-unit substrate where the original contrast reverses, a preregistered intervention confirms that replacing the mean-pass-rate plug-in by the count-law functional improves learning. The estimator defines the coefficient map; the curriculum defines the unit over which it is

averaged, and the two do not commute. Evaluate data selection with the estimator it ships on, report pass@ k from retained outcomes, and treat activity as a gate, not a law.

Reproducibility statement

The repository includes exact-enumeration tests, a one-command artifact check, structured seed-block analyses, stored protocol and verdict documents, figure derivation scripts regenerating every figure from checksummed inputs (paper/results/manifest.json), a source-locked 60-run Acrobot tournament with deterministic retained-ledger reanalysis, a one-command reproduce.sh, and a generated 54-row compact registry (curriculum_maxrl/run_registry.json) indexing the maze, Countdown and GSM8K artifacts each cited number is drawn from. Secondary records and receipt manifests remain documented in the appendix. Registration timing objects that remain external are disclosed rather than treated as locally auditable. The independent unit is stated for every central result; known raw-data gaps and secondary-study provenance are disclosed in App. D.

AI use statement

AI assistants supported planning, implementation, debugging, literature discovery, statistical and artifact auditing, figure preparation, and manuscript editing. The authors selected the questions and decision rules, ran the studies, verified derivations, code, numerical results, and citations, and take responsibility for every claim.

References

- Tajwar, F., Zeng, G., et al. Maximum Likelihood Reinforcement Learning. ICML, 2026.
- Yue, Y., et al. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? NeurIPS (oral), 2025.
- Yuan, S., Chen, J., Zheng, J., Li, M., Feng, L., Wang, D., Xiang, T., Liu, T., An, B. Understanding diversity collapse in RLVR via the lens of overtraining. arXiv:2606.15455, 2026.
- Cui, G., et al. The entropy mechanism of reinforcement learning for reasoning language models. arXiv:2505.22617, 2025.
- Cai, P., Fang, T., Li, X., Zeng, Q., Li, G., Chen, J. Curriculum reinforcement learning can incentivize reasoning capacity in LLMs beyond the base model. arXiv:2606.22317, 2026.
- Foster, T., Sims, A., Forkel, J., Foerster, J. LILO: Learning to reason at the frontier of learnability. NeurIPS, 2025.
- Tzannetos, G., Ribeiro, B. G., Kamalaruban, P., Singla, A. Proximal curriculum for reinforcement learning agents. TMLR, 2023.
- Zheng, H., Zhou, Y., Bartoldson, B. R., et al. Act only when it pays: efficient reinforcement learning for LLM reasoning via selective rollouts. NeurIPS, 2025. arXiv:2506.02177.
- Mao, Y., Qu, Y., Wang, Q., Zou, H., Ji, X. Dynamics-predictive sampling for active RL finetuning of large reasoning models. ICLR, 2026. arXiv:2603.10887.
- Nguyen, H. T., Nguyen, B., Ma, W., Zhao, Y., She, R., Nguyen, V. A. Adaptive rollout allocation for online reinforcement learning with verifiable rewards. ICLR, 2026. arXiv:2602.01601.
- Jiang, G., Feng, W., Quan, G., et al. VCRL: variance-based curriculum reinforcement learning for large language models. arXiv:2509.19803, 2025.
- Jiang, M., Grefenstette, E., Rocktäschel, T. Prioritized level replay. ICML, 2021.
- Jiang, M., Dennis, M., Grefenstette, E., Rocktäschel, T. minimax: efficient baselines for autocurricula in JAX. Agent Learning in Open-Endedness Workshop at NeurIPS, 2023.
- Dennis, M., Jaques, N., Vinitzky, E., et al. Emergent complexity and zero-shot transfer via unsupervised environment design. NeurIPS, 2020.

-
- Parker-Holder, J., Jiang, M., Dennis, M., et al. Evolving curricula with regret-based environment design. ICML, 2022.
- Portelas, R., Colas, C., Hofmann, K., Oudeyer, P.-Y. Teacher algorithms for curriculum learning of deep RL in continuously parameterized environments. CoRL, 2020.
- Wang, T., Li, S., Sun, Y., Ding, D., Dobriban, E. Where to spend rollouts: hit-utility optimal rollout allocation for group-based RLVR. arXiv:2605.07114, 2026.
- Jiang, H., Liu, H., Mirzasoleiman, B. Learning as reasoning unfolds: progressive rollout allocation for efficient reinforcement learning. arXiv:2607.22002, 2026.
- Yang, Z., Guo, Z., Huang, Y., et al. Depth–breadth synergy in RLVR: unlocking LLM reasoning gains with adaptive exploration. arXiv:2508.13755, 2025.
- Xiong, W., et al. Reinforce-Ada: an adaptive sampling framework under non-linear RL objectives. arXiv:2510.04996, 2025.
- Yang, Y., Liu, J., He, Y., Zhang, L., Schölkopf, B., Jin, Z. Transferability for general reasoning: an automated curriculum for multi-domain RLVR. arXiv:2606.25178, 2026.
- Chen, X., Lu, J., Kim, M., et al. Self-evolving curriculum for LLM reasoning. arXiv:2505.14970, 2025.
- Wang, Z., Cui, G., Li, Y.-J., Wan, K., Zhao, W. DUMP: automated distribution-level curriculum learning for RL-based LLM post-training. arXiv:2504.09710, 2025.
- Jiang, X., Tang, Z., Lin, W., et al. From reasoning chains to verifiable subproblems: curriculum reinforcement learning enables credit assignment for LLM reasoning. arXiv:2605.22074, 2026.
- Zheng, Y. RL2ML: finite-rollout surrogate objectives from reinforcement learning to maximum likelihood. arXiv:2605.30154, 2026.
- Rutherford, A., et al. No regrets: investigating and improving regret approximations for curriculum discovery. NeurIPS, 2024.
- Chen, Z., et al. Pass@ k training for adaptively balancing exploration and exploitation of large reasoning models. arXiv:2508.10751, 2025.
- Chen, M., Tworek, J., et al. Evaluating large language models trained on code. arXiv:2107.03374, 2021.
- Andrychowicz, M., et al. Hindsight experience replay. NeurIPS, 2017.
- Ghosh, D., et al. Learning to reach goals via iterated supervised learning. ICLR, 2021.
- Butt, N., et al. CodeIt: self-improving language models with prioritized hindsight replay. ICML, 2024.
- Xu, I., Jiang, S., et al. Learning more from less: reinforcement learning from hindsight. arXiv:2607.09042, 2026.
- Bay, Y. Y. GRPO, Dr. GRPO, and DAPO are three operations on one number: the group-standard-deviation identity. arXiv:2607.00152, 2026.
- Gu, Z., et al. Actor-Curator: co-adaptive curriculum learning via policy-improvement bandits for RL post-training. arXiv:2602.20532, 2026.
- Qu, Y., et al. Can prompt difficulty be online predicted for accelerating RL finetuning of reasoning models? KDD, 2026. arXiv:2507.04632.
- Cui, P. Learning-zone energy: online data selection for efficient RL post-training. arXiv:2605.17003, 2026.
- Plyusov, D., Gorbatovski, A., et al. F-GRPO: Don’t let your policy learn the obvious and forget the rare. arXiv:2602.06717, 2026.
- Zhou, T., et al. When RLVR shrinks the reasoning boundary: diagnosing pass@ k inversion. arXiv:2607.20543, 2026.
- Zhang, T., Liu, F., Wong, J., Abbeel, P., Gonzalez, J. E. The wisdom of hindsight makes language models better instruction followers. ICML, 2023.

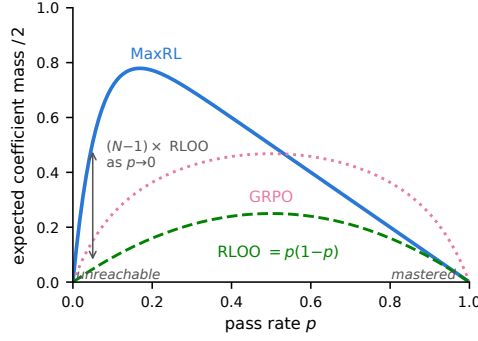


Figure 3: Three estimators, three activity geometries. Exact expected half-mass at $N = 16$ under zero-stabilizer MaxRL, gradient-averaged RLOO, and the zero-stabilizer sample-SD-normalized GRPO convention. Coefficient mass is sign-blind, so the curves motivate rather than prove the empirical coverage ordering.

A Proof details

Truncation order. For $K > 0$, conditioning on one sampled response and using exchangeability gives

$$\mathbb{E}\hat{g} = \left(\frac{1 - (1-p)^N}{p} - (1-p)^{N-1} \right) \nabla p.$$

Since $1 - (1-p)^N = p \sum_{j=0}^{N-1} (1-p)^j$, subtracting the final term leaves $\sum_{j=0}^{N-2} (1-p)^j \nabla p = \nabla J_{N-1}$. If the centered control-variate term is retained on $K = 0$, its contribution is $(1-p)^{N-1} \nabla p$. The expectation becomes $\sum_{j=0}^{N-1} (1-p)^j \nabla p = \nabla J_N$. Thus the direct/non-dropped convention has order N , while the deployed hybrid convention above has order $N-1$; theorem and implementation must use the same convention. At audited MaxRL source commit 7197bbb, the released path computes $(r_i - \bar{r})/(\bar{r} + \epsilon)$ and zeros all-failure groups; after the deployed gradient average, its $\epsilon \rightarrow 0$ coefficients match this convention up to common scale. The parsed truncation-order setting is inactive on that path.

Coefficient mass. For $0 < K < N$, each success has coefficient $(N-K)/(KN)$ and each failure has coefficient $-1/N$. The two classes each contribute $(N-K)/N$ in absolute value. For $K \in \{0, N\}$ the deployed centered estimator contributes zero. With $K \sim \text{Bin}(N, p)$,

$$\begin{aligned} A_N(p) &= \frac{2}{N} \sum_{k=1}^{N-1} (N-k) \binom{N}{k} p^k (1-p)^{N-k} \\ &= \frac{2}{N} \{ \mathbb{E}[N-K] - N \Pr[K=0] \} \\ &= 2\{1-p - (1-p)^N\}. \end{aligned}$$

The peak follows from $u'_N(p) = N(1-p)^{N-1} - 1$ and strict concavity for $N > 1$.

Posterior-integrated activity. For $p \sim \text{Beta}(a, b)$, $\mathbb{E}[p] = a/(a+b)$ and the beta-function ratio gives

$$\mathbb{E}[(1-p)^N] = \frac{B(a, b+N)}{B(a, b)} = \frac{(b)_N}{(a+b)_N}.$$

Substitution in $u_N(p) = 1 - (1-p)^N - p$ proves the deterministic posterior priority in §2. Subtracting it from $u_N(\mathbb{E}[p])$ gives the printed nonnegative Jensen gap.

Estimator limits. Under the deployed $1/N$ gradient-average normalization, RLOO expected half-mass is $p(1-p) = u_2(p)$, so $u_N(p)/u_2(p) \rightarrow N-1$ as $p \rightarrow 0$. Near mastery, both are first order in $q = 1-p$. With sample-SD normalization, GRPO group mass

Table 2: Contribution–evidence map. Status is attached to the specific object evaluated, not transferred across rows.

Object	Claim/mechanism	Evidence allowed here
Coefficient identity	Exact A_N , $T = N - 1$, and posterior-integrated u_N for the stated practical-MaxRL convention	Proved; estimator activity only, not learning progress
Direct teacher	fixed-pool Thompson-scored u_N with proportional sampling and a uniform floor	Scoped empirical; Acrobot supports the score in one fixed family, while Digits rejects a universal mapping
Coefficient-Activity PLR	Analytic posterior activity replaces MaxMC priority inside robust PLR; replay and staleness remain PLR mechanisms	Development negative (§3.3); as a replacement for MaxMC it does not beat upstream robust PLR on AMaze at 5,000 updates

is proportional to $\mathbb{E}[\sqrt{K(N - K)}]$ up to deployed normalization constants; its relative tail scaling is verified by exact enumeration in `verify_guidance_math.json`. These are coefficient statements, not gradient-direction statements.

Dependent rollouts. The per-group identity does not require independence. For any binary group,

$$\mathbb{E} \sum_i |w_i| = 2\{\Pr(K \geq 1) - \mathbb{E}[K]/N\}.$$

With a common marginal p , the second term is p ; independence is used only to replace $\Pr(K \geq 1)$ by $1 - (1 - p)^N$. In a post-guidance exact beta-binomial CPU stress test, count-distribution enumeration matched the general identity to 2.58×10^{-13} . At $\rho = .10$, the activity peak moved from .169 to .234 for $N = 16$ and from .106 to .186 for $N = 32$. This dependence model shows how increasing positive correlation can lower activity inside that family; pairwise correlation alone does not determine a general joint law. The result is a scope diagnostic, not evidence that coefficient mass causes learning or that beta-binomial dependence describes neural rollouts.

Posterior-mean bias and the uncertainty penalty. With $p_x \mid \mathcal{D} \sim \text{Beta}(a_x, b_x)$, strict concavity of u_N for $N > 1$ and Jensen’s inequality give $\mathbb{E}[u_N(p_x) \mid \mathcal{D}] \leq u_N(\mathbb{E}[p_x \mid \mathcal{D}])$, so a priority computed from the posterior mean overstates expected activity. The gap is nonnegative and closed-form,

$$u_N\left(\frac{a_x}{a_x + b_x}\right) - \mathbb{E}[u_N(p_x) \mid \mathcal{D}] = \frac{(b_x)_N}{(a_x + b_x)_N} - \left(\frac{b_x}{a_x + b_x}\right)^N,$$

and vanishes as the posterior concentrates. It is an uncertainty penalty, not an exploration bonus: it shrinks the priority of tasks whose pass rate is still poorly determined.

B Algorithms and estimator contracts

Posterior teacher. Each task stores discounted successful and failed rollout counts. A Thompson draw $\tilde{p}_x$ is mapped through u_N and mixed with a uniform floor. Teacher state, sampler cursor, and checkpoints restore together. Observations use pre-relabel rewards so recycled outcomes do not inflate the requested task’s posterior.

Coefficient-Activity PLR (development negative). The AMaze implementation is a narrow overlay on robust PLR in the pinned minimax substrate (Jiang et al., 2021; 2023). It uses cumulative Beta evidence per replay-buffer slot and replaces MaxMC priority with deterministic $\mathbb{E}[u_N(p) \mid \mathcal{D}]$; the declared N may equal `n_eval` (exact grouping) or be decoupled from it. Buffer admission, the robust replay gate, inverse-rank mass, staleness mixing, and with-replacement draws remain PLR mechanisms. Tie-aware ranking is shared with the matched MaxMC control. §3.3 reports the development sweep: the replacement priority does not beat upstream robust PLR, and no baseline-performance claim is made.

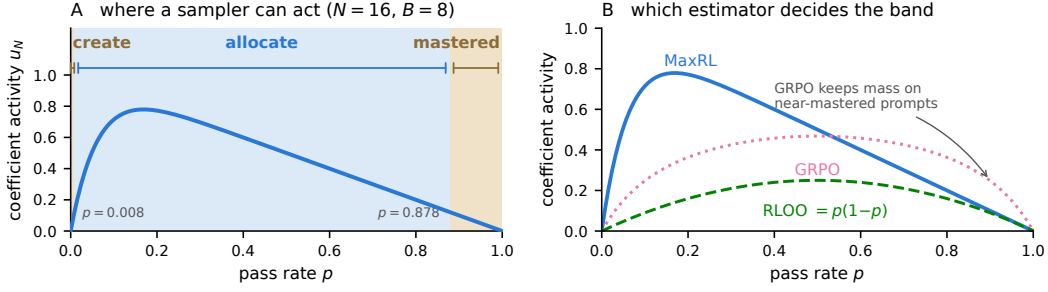


Figure 4: The three-regime map: the teacher allocates, hindsight creates, the estimator decides whether either is safe. (A) Coefficient activity has exact zeros at $p=0$ and $p=1$, so the pass-rate axis splits into a band where a sampler can redistribute existing signal and two regions where no priority rule can help. Edges are budget-dependent: with B groups per step a task is operationally dead at budget when it yields under one live group, $B\{1 - (1 - p)^N\} < 1$; drawn at $B=8$, $N=16$. Relabeling is the only channel that adds signal outside the band (App. C). (B) Which estimator is deployed decides the band’s shape. GRPO’s geometry retains mass on near-mastered prompts, so a frontier curriculum is not automatically safe under it.

Relabeling (secondary). For an all-fail request, a verified achieved goal can create an auxiliary update where repeating the request would emit none. A parser proposes that goal from a failed response, and the environment’s exact verifier must accept the response under that goal. Prompt conditioning and explicit response goal declarations are rewritten; answer spans, arithmetic, decimals, signed values, and unknown contexts are preserved. The reported-run implementation is frozen, while a conservative release wrapper and 26 CPU regression/property tests protect future runs. Relabeled groups remain auxiliary selected-data updates; no importance ratio converts them into fresh on-policy target samples. The intervention changes both update dose and direction, and its achieved-goal distribution is biased toward outputs the policy already reaches (Andrychowicz et al., 2017; Ghosh et al., 2021).

C Experimental protocols and secondary results

The creation channel, and how to gate it. Where activity is zero no sampler can help, and relabeling a failed rollout as a success of the goal it did reach is the only channel that adds signal (Fig. 4A). Two results carry to practice. Gating relabels by an estimated destination pass rate rather than by observed frequency is registered and confirmed: one probe rollout per step recovers 98% of the frequency-to-oracle gap (AUC .798 $\rightarrow$.877 against an oracle .879; 10/10 seeds), and probe budgets of 1, 4 and 16 are indistinguishable — P-PB1 confirmed, while its monotonicity sub-hypothesis P-PB2 is not supported (1/10 seeds). Descriptively, how relabels are grouped matters as much as whether they happen: each relabeled row as its own $K=1$ group scores .952, one destination per group .881, and a shared- K coupled form .749, against .705 with no recycling — both orderings 10/10 seeds, ten-seed AUC means on the fixed chain pool, exploratory.

Fixed-completion N -sweep (post-guidance, descriptive). 36-task CPU skill-chain testbed, $N \in \{2, 4, 8, 16, 32\}$, eight paired seeds, no recycling, exactly 51,200 sampled completions per cell, teacher decay normalized per completion so changing N does not change the memory horizon. Against uniform the contrasts are $-.0106$, $-.0032$, $+.0309$, $+.0453$, $+.0836$; positive-seed counts against $p(1-p)$ are identical at $N=2$ then 8/8, 8/8, 8/8, 8/8, and against uniform 3/8, 3/8, 8/8, 7/8, 7/8. The $N=2$ identity reproduces bit-for-bit. Designed after the theoretical guidance existed, hence not preregistered, and on one co-designed testbed: evidence for rollout awareness, not a scaling law or a blanket curriculum advantage.

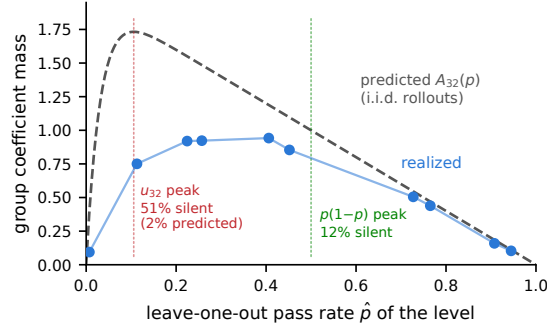


Figure 5: The task-granularity gap (descriptive, computed after the frozen primary). Plug-in prediction evaluates A_{32} at a level’s mean pass rate, estimated leave-one-out from the other groups on that level so mass is never binned against its own K . Realized mass tracks the predicted shape ($r = .90$, 288,000 group draws) but lies below it by exactly twice the level’s excess all-fail probability (Cor. 2), a gap that widens as p falls.

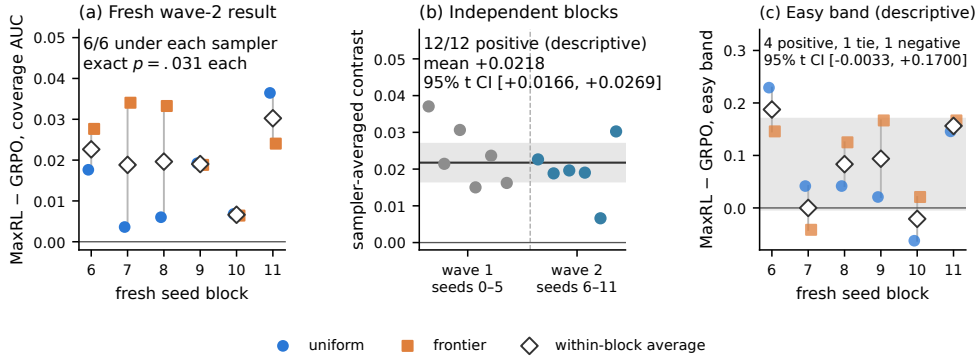


Figure 6: Maze factorial at the independent seed-block unit. (a) The externally recorded fresh-wave result is positive in 6/6 blocks under each sampler (exact sign $p = .03125$ each). Connected observations share a block. (b) Sampler-averaging gives wave-2 mean $+0.0218$ and post-hoc 95% CI $[+0.0166, +0.0269]$; 12/12 cross-wave block averages are descriptive. (c) Easy-band localization is suggestive: four positive blocks, one tie, one negative, and an interval containing zero.

Fixed-completion N -sweep. The environment contains three nested 12-level chains and ten actions. Every cell runs exactly 51,200 completions, with arms for uniform, $p(1 - p)$, and u_N sampling under the practical MaxRL estimator, no hindsight, and eight paired seeds. Checkpoints share a completion clock. The post-guidance analysis is descriptive.

Digits factorial (internally frozen negative). On a stored 1,077/360/360 train/development/test split of sklearn Digits, a 4,810-parameter float64 policy recomputes each example’s exact correct-class probability before every update. We crossed practical MaxRL and RLOO with uniform, $p(1 - p)$, and u_8 sampling. Five-rate development selected learning rate .1 for MaxRL, RLOO, and the common-rate check. Across 24 fresh paired blocks, the internally frozen interaction was $+0.01589$ (paired bootstrap 95% CI $[-0.01686, +0.04712]$, exact two-sided sign-flip $p = .350$; 15/24 positive), so it was not supported despite passing the treatment-delivery gate. MaxRL did favor u_8 over $p(1 - p)$ by $+0.20842$ (CI $[+0.16791, +0.24744]$; 23/24), but RLOO’s frozen prediction reversed: $p(1 - p) - u_8 = -0.17665$ (CI $[-0.22042, -0.13593]$; 0/24). Moreover, the matched samplers were below uniform for both MaxRL (-0.11279) and RLOO (-0.37581). Because every selected rate equaled .1, the tuned and common schedules have byte-identical ledgers and checkpoints and identical scientific summary content after removing phase/authorization

labels; they are identity checks, not independent replications. This exact- p contextual bandit therefore supports neither the planned interaction nor a universal curriculum advantage; it shows why expected coefficient mass must remain an estimator-activity proxy rather than a learning law. Its internally hashed local lock is not backed by an immutable public pre-execution commit that independently establishes timing.

Acrobot V3 (historical). The frozen control uses official Acrobot-v1, eight nested thresholds from -1.5 through native success at 1.0 , one shared H64 actor, practical MaxRL, $N = 16$, and no hindsight. Uniform and u_{16} receive 20 paired seeds and a nominal two million actual transitions per run, completing the final group. Evaluation uses 32 shared trajectories total, reused across all eight predicates, at initialization and each 100,000 transitions with fixed common random numbers. The internally frozen primary is target-uniform normalized transition-AUC; group diagnostics and evaluation curves for all 40 runs are vendored. A post-hoc audit found cross-seed overlap between actor-initialization and action-sampling RNG roots, so the retained effect and mechanism audit are descriptive rather than clean confirmation. The mechanism audit is deterministic from those records.

Acrobot fixed-pool tournament V2. The replacement reruns uniform and u_{16} and adds $p(1-p) = u_2$ on fresh logical seeds 20000–20019. All 60 source/runtime-locked runs completed after an outcome-blind development gate; globally disjoint RNG roots and per-run group-diagnostic and aggregate-checkpoint ledgers were retained. The primary $u_{16} - p(1-p)$ transition-AUC difference is $+.04803$ (paired bootstrap 95% CI $[+.02094, +.07385]$, exact sign-flip $p = .003361$), clearing the predeclared $+.01$ estimate filter; Table 1 gives the Holm-controlled uniform comparisons. Observed mass means are close between adaptive arms ($.7344$ versus $.7490$ per group; 115.57 versus 115.90 per million transitions), both descriptive paired intervals include zero, and no equivalence test was frozen. The p -value tests zero rather than the SESOI, and pairing does not guarantee sign exchangeability. The $p(1-p)$ arm isolates only the score on our scaffold, not full ProCuRL or SFL.

Paid-probe ProCuRL selection semantics. This is a separate 80-pair family, not an extension of the V2 table. It keeps the same eight thresholds, H64 actor, practical MaxRL estimator, $N = 16$, and 3×10^{-4} learning rate. Its four arms are ordinary uniform; a uniform sham with paid probes; source-faithful ProCuRL selection softmax $\{20\hat{p}(1-\hat{p})\}$; and continuous-range-matched u_{16} selection. Probed arms run 20 episodes per task initially and at every crossed 5,120-student-transition boundary; all student and probe transitions count toward the nominal two-million paid budget. Across seeds 21000–21079, the internally frozen fixed-paid-AUC contrast u_{16} –ProCuRL is $+.004894$ (paired SD $.022139$, $t_{79} = 1.9773$, $p = .05149$; 20,000-resample CI $[+.000110, +.009727]$; one-million sign-flip $p = .05161$). It misses both the $.02$ SESOI and the $.05$ paired- t criterion. The five Holm-family contrasts are ProCuRL–sham $-.001704$ (adjusted $p = .4568$), u_{16} –sham $+.003190$ ($.4547$), ProCuRL–ordinary $-.313774$ (3.47×10^{-66}), u_{16} –ordinary $-.308880$ (9.48×10^{-67}), and sham–ordinary $-.312070$ (4.06×10^{-68}). Mean fixed-paid AUCs are $.33771$ (ProCuRL), $.33942$ (sham), $.65149$ (ordinary), and $.34261$ (u_{16}).

The probed arms average 28.0–28.1 sweeps and 140.5–141.1k student transitions versus 2.003M for ordinary uniform; 93.19–93.21% of their paid transitions are probes. They average 15.4–19.8 optimizer updates versus 259.1 ordinary-uniform updates, with mean paid-budget overshoot 69,960–72,939 transitions versus 2,984. All 320 confirmation and 12 outcome-blind development runs reconcile exactly, all 11 development gates pass, and the common-random-number invariants replay.

An earlier outcome-blind development wave was invalidated before any gate or arm contrast because 73 entropy-sum replays differed by at most 7.28×10^{-12} . Its seeds were burned and its incident receipt is retained; the reported family uses fresh seeds and a replacement lock.

The external confirmatory raw is 1,374,886,097 bytes. Its SHA-256 boundary is:

```
raw: b1f8756c249effab8c77101c8bca73ddf708a5e143c18fe8742fd5712 added7c12
lock: b7c7f76faaffa1fe65557717bfe545f2ec850495d370cd97fe72a8871fe8d0f
analysis: 2010e30b5b15a212e2d6bdfaacd43d2434e5f468a96be613cf59744b9bc2fb38
```

The source bundle carries a 320-record external-raw manifest: compact mode checks its small-artifact bindings, while full verification requires the raw file and replays locked validation. The local lock is mechanical provenance, not an immutable public pre-execution registration.

Post-guidance rollout-allocation factorial. We also compared fixed $N = 16$, published-HORA-style posterior hit allocation (Wang et al., 2026), and a mass-aware marginal that counts four already-spent probe rollouts. A 2-sampler by 3-allocator CPU factorial uses 16 paired seeds and 51,200 completions per cell. The mass-aware arm improves pass@8 AUC over HORA-style allocation by $+0.00676$ (11/16 positive pairs) and over fixed groups by $+0.02517$ (14/16). However, it reduces realized coefficient mass per completion by $.000306$ relative to HORA-style allocation in all 16 pairs, falsifying the frozen mechanism prediction. Its group sizes are also extreme (cell-average maximum about 76 versus 57 for HORA and 16 fixed). We therefore treat the result as exploratory motivation for posterior-calibration and allocation-cap ablations, not a fourth contribution.

A frozen follow-up crossed two samplers, the two adaptive scores, four caps, and three information sources over 16 paired seeds (50 cells; 800 runs). Restricting to the 32 deployable-information adaptive cells, all 32 adaptive-minus-fixed mean pass@8-AUC contrasts were positive, with 22 unadjusted descriptive Student- t intervals above zero; no multiplicity-adjusted inference is assigned. Yet realized coefficient mass was lower in 28/32 cells, with 24 intervals below zero and none above zero. The remaining 16 oracle-preupdate cells are diagnostic. Thus the expanded result reinforces the learning-mass mismatch rather than a mediation claim. The frozen engineering rule chose cap 32: averaged over samplers for the fresh-group/same-step cells, it reduced maximum group size by 58.07% with an AUC change of $-.00271$ versus uncapped. This is a point-estimate filter for a future pilot, not an equivalence test. The source/runtime lock is internally hashed but lacks a public pre-execution commit establishing timing.

Maze factorials. Wave 1 has six shared seed/warm-start blocks and six arms per block: MaxRL and GRPO under uniform and FrontierMax sampling plus two diagnostic GRPO arms. Its recorded endpoint hypothesis failed. Wave 2 has six fresh blocks and the four core arms. The external execution record names time-integrated pass@8 relative to warm start primary, with at least 5/6 positive blocks required under each sampler. The block analyzer reports sampler-specific exact sign tests, sampler-averaged contrasts, post-hoc intervals, and leave-one-block-out checks. No pooled p -value treats sampler observations as independent.

AUC robustness status. The frozen multiverse covers integration rule, warmup inclusion, horizons, and leave-one-checkpoint-out variants, but all 24 wave-2 checkpoint JSONLs and warm starts remain external. Structured summaries recover the anchors (uniform 6/6, mean $+0.01496$; FrontierMax 6/6, $+0.02404$; sampler-averaged 6/6, $+0.01950$), but cannot identify those variants; the analyzer therefore exits explicitly blocked.

Countdown. We fine-tune SmolLM2-360M with $N = 16$. The v2 pool uses operand permutation, parenthesization, and exact division. Tier 1 has 128 evaluation tasks and 16 verifier-scored samples per task. B1 disables recycling; B2 admits exact-verifier achieved goals; the committed scoreboard contains three-seed aggregates. ARM B disables recycling and uses two optimizer epochs on every live group; its three individual endpoints are vendored. Headline endpoints are step-60 mean@16 and VERL bootstrap best@16. The latter uses 1,000 with-replacement resamples of size 16 per prompt and is not standard unbiased pass@16; missing per-task outcomes prevent recomputation. A stored task-identity summary reports zero tier-1 and tier-2 SFT overlap and 27 exposed tier-0 tasks; the source manifests needed to recompute those counts are absent. Tier 0 is excluded from absolute claims.

The reported recycling-package aggregate moves tier-1 mean@16 from $.278 \pm .066$ to $.324 \pm .014$ while bootstrap best@16 moves from $.541 \pm .024$ to $.492 \pm .013$ (mean $\pm$ sample SD across three seeds). Complete three-seed B1/B2 records are not vendored, so this supports neither

a paired seed test nor a named-method effect claim. It is retained only as a measurement caution: mean accuracy alone cannot reveal whether coverage was spent to buy it.

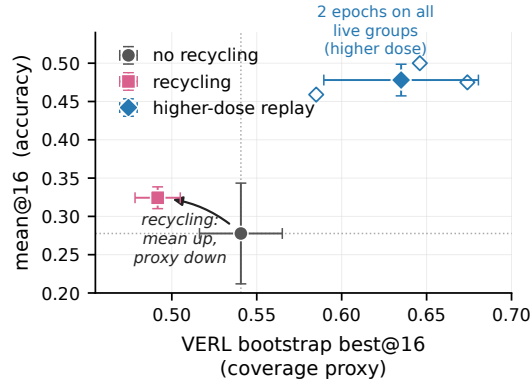


Figure 7: Secondary Countdown accuracy-coverage-proxy record. Error bars are sample standard deviations across three seeds; seed 3 for B1/B2 is aggregate-only. The recycling-package aggregate has higher mean@16 and lower VERL bootstrap best@16. The live-group replay control uses two optimizer epochs on every live group, so it cannot separate dose from update direction.

The externally specified replay control reaches mean@16 $.478 \pm .021$ and bootstrap best@16 $.635 \pm .046$, with all three endpoints vendored. Recycling permits at most 12 auxiliary groups per 64 requested groups (18.75%; realized dose can be lower), whereas replay doubles optimizer epochs on every live group. This higher-dose alternative improves both logged metrics without identifying a dose-monotonic causal bound.

Negative branches. The artifact retains the failed maze endpoint, gate, Jugs, and robotics branches. GSM8K missed its delivery threshold (.601480 versus $< .60$) and is inconclusive; the single-seed Countdown per- k curve is exploratory. The corrected-code gate dose-response (frozen protocol, three paired seeds, independent A100 environment, retained 16-bit raw outcomes per task) passed its transfer gate — ungated recycling beat no-recycling on mean@16 in 3/3 seeds — and its manipulation check (rejection graded 0/.72-.80/.88-.90/.93 across gate strengths), but neither tested setting met the frozen usefulness criterion; per-seed signs were mixed at $n = 3$. Its raw outcomes also record, descriptively, one seed in which mean@16 rose while standard observed-set pass@16 fell $.656 \rightarrow .414$ — a non-proxy instance of the ambiguity the operational recommendation targets.

D Artifact and data accounting

bash reproduce.sh -build tests the identities and integrations, verifies declared manifest inputs and the local compact registry, runs stored endpoint spot checks, regenerates figures, and builds the canonical main_iclr2027.pdf plus the extended main.pdf. The compact registry has 54 records: 35 maze, 11 Countdown, 7 GSM8K, and one P0 analysis artifact. The broader 562-row registry used during release-branch auditing is a different object; it is not copied over this checkout’s registry and is not used for the local count.

Missing external data are maze checkpoints; Countdown task manifests, outcomes, and B1/B2 seed 3, so the clean 101-task tier-0 analysis cannot be reconstructed; the 1.375 GB paid-probe confirmatory raw (its content hash and per-run receipt manifest ship); and the Digits 5.08 GB replay payload (block contrasts and a content manifest ship). Compact inputs and checksums are in paper/results/manifest.json and fail closed.